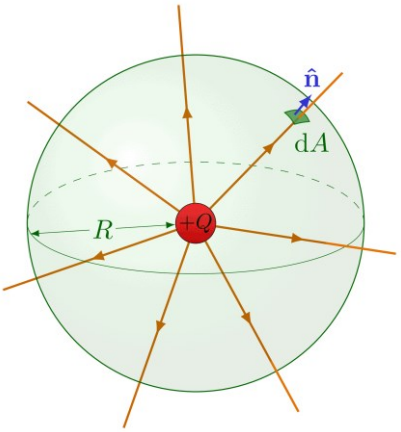


XIAMEN UNIVERSITY MALAYSIA

EP2T #5

Gauss's Law

Lecture 5



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Question 1:

A flat sheet is in the shape of a rectangle with sides of lengths 0.400 m and 0.600 m. The sheet is immersed in a uniform electric field of magnitude 90.0 N/C that is directed at 20° from the plane of the sheet (**Fig. T5.1**). Find the magnitude of the electric flux through the sheet.

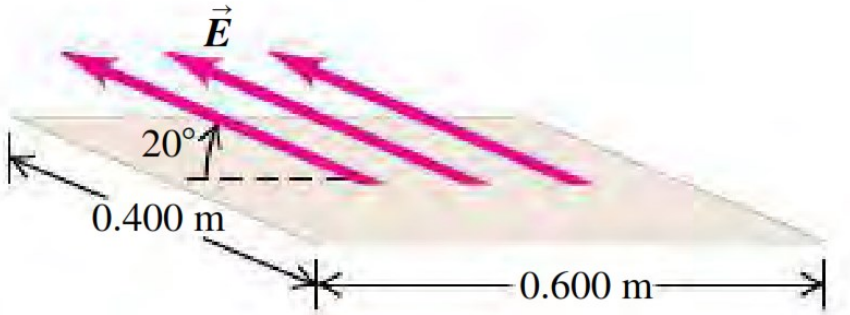
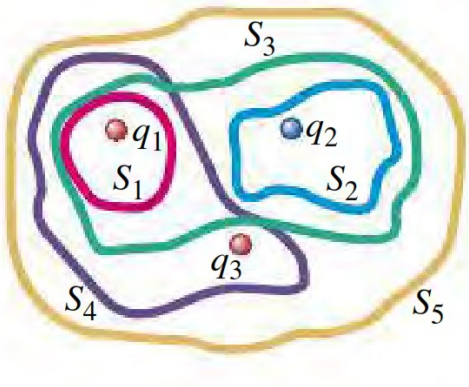


Figure T5.1

Question 2:

The three small spheres shown in **Fig. T5.2** carry charges $q_1 = 4.00$ nC, $q_2 = -7.80$ nC, and $q_3 = 2.40$ nC. Find the net electric flux through each of the following closed surfaces shown in cross section in the figure: (a) S_1; (b) S_2; (c) S_3; (d) S_4; (e) S_5. (f) Do your answers to parts (a)–(e) depend on how the charge is distributed over each small sphere? Why or why not?



Surface	What it encloses
S_1	q_1
S_2	q_2
S_3	q_1 and q_2
S_4	q_1 and q_3
S_5	q_1 and q_2 and q_3

Figure T5.2

Question 3:

The electric field 0.400 m from a very long uniform line of charge is 840 N/C. How much charge is contained in a 2.00-cm section of the line?

Question 4:

An electron is released from rest at a distance of 0.300 m from a large insulating sheet of charge that has uniform surface charge density $+2.90 \times 10^{-12}$ C/m^2. (a) How much work is done on the electron by the electric field of the sheet as the electron moves from its initial position to a point 0.050 m from the sheet? (b) What is the speed of the electron when it is 0.050 m from the sheet?

Question 5:

A conductor with an inner cavity, like that shown in Fig. T3.3 , carries a total charge of +5.00 nC. The charge within the cavity, insulated from the conductor, is −6.00 nC. How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

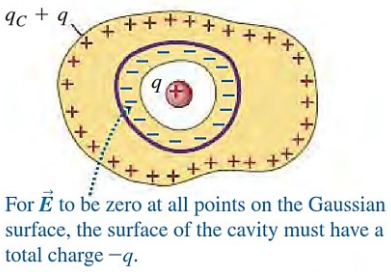


Figure T3.3 An isolated charge q placed in the cavity

Question 6:

A very large, horizontal, nonconducting sheet of charge has uniform charge per unit area $\sigma = 5.00 \times 10^{-6}\ \mathrm{C/m^2}$. (a) A small sphere of mass $m = 8.00 \times 10^{-6}$ kg and charge q is placed 3.00 cm above the sheet of charge and then released from rest. (a) If the sphere is to remain motionless when it is released, what must be the value of q? (b) What is q if the sphere is released 1.50 cm above the sheet?

Question 7:

A square insulating sheet 80.0 cm on a side is held horizontally. The sheet has 4.50 nC of charge spread uniformly over its area. (a) Calculate the electric field at a point 0.100 mm above the center of the sheet. (b) Estimate the electric field at a point 100 m above the center of the sheet. (c) Would the answers to parts (a) and (b) be different if the sheet were made of a conducting material? Why or why not?

Question 8:

At time $t = 0$ a proton is a distance of 0.360 m from a very large insulating sheet of charge and is moving parallel to the sheet with speed 9.70×10^2 m/s. The sheet has uniform surface charge density 2.34×10^{-9} C/m^2. What is the speed of the proton at $t = 5.00 \times 10^{-8}$ s?

- END -

Extra Questions for practice:

Question 9:

A very small object with mass 8.20×10^{-9} kg and positive charge 6.50×10^{-9} C is projected directly toward a very large insulating sheet of positive charge that has uniform surface charge density 5.90×10^{-8} C/m^2. The object is initially 0.400 m from the sheet. What initial speed must the object have in order for its closest distance of approach to the sheet to be 0.100 m?

Question 10:

In a region of space there is an electric field $\vec{E}$ that is in the z-direction and that has magnitude $E = [964\ \text{N/(C}\cdot\text{m)}]x$. Find the flux for this field through a square in the xy-plane at $z = 0$ and with side length 0.350 m. One side of the square is along the $+x$-axis and another side is along the $+y$-axis.

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